

B.TECH | BCSE497J — PROJECT I

ARIA

Autonomous Reasoning & Interaction Agent

A cognitive, multi-agent vision-language-action software stack for adaptive manipulation planning

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Aim

To design and simulate ARIA — a locally-run, multi-agent vision-language-action system that lets an affordable 5-DoF robotic arm understand natural-language commands, perceive objects using only a monocular camera, plan affordance-aware grasps, and execute reliable pick-and-place manipulation, while continuously monitoring its own confidence, safety, and performance.

Abstract

- ARIA is a locally-run, multi-agent manipulation system built around an affordable 5-DoF arm. It understands spoken or typed instructions, works through them step by step, and picks up, moves, and manipulates objects roughly the way a person would — adjusting for uncertainty, checking in with the user when unsure, and steadily improving its own skills over time.
- Perception is deliberately hybrid: YOLO detection combines with monocular depth (Depth-Anything / MiDaS) and camera-table calibration geometry to recover a reliable Z-coordinate, with no dedicated depth sensor anywhere in the setup. Grasp candidates from GraspNet / AnyGrasp are filtered by an Affordance Agent, and a local LLM Planning Agent orchestrates roughly fifteen specialist agents, pausing for user approval whenever its confidence drops below threshold.

Literature Review — Gaps & How ARIA Differs (1/2)

Paper	Gap Identified	What ARIA Does Differently
π 0.5 (Physical Intelligence, 2025)	Needs large-scale training data / compute; no built-in human-in-the-loop safeguard	Modular, lightweight pipeline on a single consumer GPU with confidence-gated user approval instead of pure end-to-end learning
SMART-LLM (IROS 2024)	Plans tasks but has no perception, affordance, or safety grounding of its own	LLM Planning Agent is wired directly to Vision, Affordance, Reachability, and Safety agents via function-calling
Depth Anything V2 (2024)	Monocular depth is relative, not metric — unusable alone for real-world pick coordinates	Combined with camera-table calibration geometry to recover a validated, metric Z-coordinate with no depth sensor
AffordGrasp (IROS 2025)	Affordance reasoning evaluated in isolation; no fallback when the model is uncertain	Affordance Agent output is coupled to a confidence-gated approval step before any uncertain grasp executes
SAM 2 (Meta AI, 2024)	Produces masks only — no downstream grasp planning or use-case reasoning	Segmentation feeds straight into affordance-based grasp-region ranking, not just bounding-box grasping

Literature Review — Gaps & How ARIA Differs (2/2)

Paper	Gap Identified	What ARIA Does Differently
GraspNet / AnyGrasp	Ranks grasps by reachability/geometry only — no notion of how an object is actually used	Affordance Agent re-ranks candidates by use-case (a cup by its handle, a tool by its shaft), not the nearest surface
CuRobo (ICRA 2023)	General-purpose collision-free planner; not tuned for a hobby-servo arm's trailing cables	Motion planning is explicitly cable-aware and self-calibrating for low-cost SG90/MG995 hardware
FoundationPose (CVPR 2024)	Strong 6D pose but compute-heavy and reliant on CAD/reference models for novel objects	Runs a lighter X/Y/Z pipeline now, with 6D pose kept as a scoped future addition rather than a hard dependency
OpenVLA / Octo	Generalist policies act as a black box — decisions aren't easily auditable	Every action carries a confidence score and an explicit chain-of-thought log, so each decision is traceable
SayCan ("Do As I Can, Not As I Say", 2022)	Grounds language in affordances but assumes existing robot infrastructure and no safety loop	Full stack — Safety Agent, Health Monitor, and confidence gating — built around \$150–200 hobby hardware

Research Gap

- Most published manipulation stacks assume a dedicated RGB-D or LiDAR sensor; few recover a reliable Z-coordinate from a plain \$10–30 webcam using monocular depth combined with calibration geometry alone.
- Multi-agent robotics research rarely attempts an agent hierarchy at ARIA's scale — around fifteen cooperating agents — running together on a single consumer GPU paired with sub-\$200 hobby-servo hardware.
- Affordance-aware grasp selection and confidence-gated human approval are typically studied in isolation; ARIA couples the two into one live decision loop.
- Explainable, chain-of-thought decision logging alongside cable-aware, self-calibrating motion planning is uncommon at this price point, limiting transparency in low-cost manipulation research.

Objectives

- Model and simulate a 5-DoF SG90/MG995 arm with a validated YOLO-based perception baseline.
- Recover reliable Z-coordinates via monocular depth and calibration geometry, without any dedicated depth sensor.
- Build an Affordance Agent that filters GraspNet / AnyGrasp candidates by real-world use-case rather than nearest surface.
- Benchmark inverse-kinematics solvers and integrate MoveIt2 for collision-free trajectory execution.
- Deploy a local LLM Planning Agent that orchestrates specialist agents through function-calling.
- Implement confidence-gated decision-making, a semantic World Model, and self-supervised skill improvement.

Identified SDGs: SDG 9 (Industry, Innovation & Infrastructure), SDG 4 (Quality Education) | **Target Outcome:** Conference / Scopus Publication

References

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